

8. [Maximum mark: 7]

Malik owns 15 recipe books which he bought while travelling.

- (a) Find the number of different ways of arranging the books in a line on a shelf. [1]

Malik's books were bought in three continents.

Six books are from Asia, five from Europe and four from Africa.

Malik decides to arrange the books on the shelf so that the books from each of the continents are grouped together.

- (b) Determine the number of different ways that Malik can do this. [3]

Malik chooses four books, all from the same continent.

- (c) Determine the number of different choices Malik could make. [3]

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